AIM-RGL: Agentic MI-CBT Framework with Reward-Guided Lookahead Search for Managing Client Resistance

by

Lu Htoo Kyaw

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Examination Committee:	Prof. Chaklam Silpasuwanchai (Chairperson)
	Prof. Chantri Polprasert
	Prof. Attaphongse Taparugssanagorn

Nationality:	Myanmar
Previous Degree:	Bachelor of Science in Information Technology
	University of Information Technology, Yangon
	Republic of the Union of Myanmar
Scholarship Donor:	AIT Scholarship

Asian Institute of Technology
School of Engineering and Technology
Thailand
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AUTHOR'S DECLARATION

I, Lu Htoo Kyaw, declare that the research work carried out for this thesis was in accordance with the regulations of the Asian Institute of Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the thesis including final revisions.

Date: July 18, 2026

Name: Lu Htoo Kyaw

Signature:

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ABSTRACT

The growing demand for psychological support has increased interest in using large language models (LLMs) as accessible counseling agents. However, therapeutic dialogue requires more than generating a locally appropriate response. In counseling, especially when clients show resistance or ambivalence, the timing of therapeutic strategies is critical. Motivational Interviewing (MI) is useful for supporting readiness, autonomy, and exploration of ambivalence, while Cognitive Behavioral Therapy (CBT) provides structured techniques for identifying and modifying unhelpful thoughts and behaviors. Introducing CBT too early may increase resistance, while remaining only in MI may slow progress when the client is ready for structured intervention.

Existing LLM-based counseling systems have explored therapeutic strategy selection, memory-based adaptation, and MI-CBT dialogue support. However, many approaches remain limited in their ability to anticipate the immediate downstream effects of a therapeutic response before it is selected. As a result, a system may choose a response that appears suitable in the current turn but does not support the client's next therapeutic state, readiness, or engagement.

This thesis proposes AIM-RGL, an agentic MI-CBT framework with reward-guided lookahead search for managing client resistance in simulated therapeutic conversations. The framework uses multiple specialist therapist agents, including open-ended questioning, reflection, affirmation, summarization, and CBT agents, to generate candidate responses. A client simulator then produces possible future client reactions based on cognitive conceptualization and stage-of-change information. A reward model evaluates candidate responses according to therapeutic dimensions such as alliance, empathy, collaboration, CBT skill, MI quality, and client readiness. Instead of selecting only the best immediate response, AIM-RGL performs short-horizon lookahead search to choose the response expected to lead to a better near-future therapeutic state.

The study evaluates whether reward-guided lookahead improves MI-CBT response quality compared with baseline approaches. It further examines how different lookahead depths, reward model backbones, and policy model backbones affect therapeutic response generation. Evaluation focuses on simulated therapist-client dialogues using metrics related to therapeutic alliance, MI fidelity, CBT competence, change talk, and

sustain talk. This thesis does not claim to measure real clinical treatment outcomes or replace professional therapists; rather, it investigates how future-aware response selection can improve the design of LLM-based counseling support systems. Overall, AIM-RGL contributes a structured framework for proactive, context-sensitive, and therapeutically grounded response generation in AI-assisted mental health dialogue.

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CHAPTER 1
INTRODUCTION

1.1 Background of the Study

Mental health support has become an increasingly important global concern as psychological distress, anxiety, depression, and related conditions continue to affect people's daily functioning, relationships, work, and quality of life. The World Health Organization reported that more than one billion people are living with mental health disorders, with anxiety and depression among the most common conditions worldwide World Health Organization (2025). Despite increasing awareness of mental health issues, access to professional psychological care remains limited, particularly in low- and middle-income countries where treatment gaps are still high Kazdin (2019); World Health Organization (2025). These challenges have encouraged researchers to explore scalable forms of psychological support, including digital mental health interventions and large language model-based counseling systems. Recent studies suggest that large language models may support mental health-related tasks, but their use in counseling requires careful evaluation because therapeutic dialogue involves empathy, safety, contextual understanding, and professional boundaries Hua, Na, Li, et al. (2025); Nguyen et al. (2025).

Large language models have shown potential for producing fluent and empathetic responses that fit counseling-like contexts. However, therapeutic communication is not only a matter of generating a warm or natural reply. A helpful response also needs to preserve trust, respect the client's autonomy, and support a collaborative relationship between the client and the therapist. These qualities are central to client-centered counseling and therapeutic alliance, where empathy, agreement on goals, and agreement on therapeutic tasks influence the quality of the interaction Bordin (1979); Rogers (1957). For LLM-based counseling systems, this means that response generation should be grounded not only in language quality, but also in therapeutic principles and interactional sensitivity.

This thesis focuses on simulated therapeutic dialogue involving resistance and ambivalence. Motivational Interviewing (MI) and Cognitive Behavioral Therapy (CBT) provide two complementary foundations for this setting. MI emphasizes collaboration, autonomy, reflective listening, affirmation, and eliciting the client's own reasons for change, making it useful when the client is uncertain or not yet ready to engage deeply with change Bischof, Bischof, and Rumpf (2021); Miller and Rollnick (2013). CBT provides structured methods for identifying unhelpful thoughts, examining evidence, developing more balanced interpretations, and supporting behavioral change Beck (2020); Chand, Kuckel, and Huecker (2023). Together, MI and CBT offer a useful combination for supporting clients who may move between hesitation, exploration, and readiness for more structured therapeutic work.

The integration of MI and CBT has also been explored in conventional mental health treatment outside the LLM context. Arkowitz and Westra discussed the use of MI with CBT for depression and anxiety, especially where resistance and ambivalence affect treatment engagement Arkowitz and Westra (2004). In generalized anxiety disorder, studies have examined MI as a pretreatment or integrated component of CBT, with findings suggesting potential benefits for clients with higher severity, resistance, or ambivalence Westra, Arkowitz, and Dozois (2009); Westra, Constantino, and Antony (2016). Reviews of MI as an adjunct to CBT for anxiety disorders also suggest that MI-CBT can support treatment initiation, engagement, and clinical outcomes, although the literature still points to the importance of how and when MI is integrated with CBT Marker and Norton (2018); Randall and McNeil (2017). This clinical background supports the motivation of this thesis: the challenge is not only combining MI and CBT, but coordinating them at the right moment in the therapeutic dialogue.

The difficulty of coordinating MI and CBT can be understood through the client's readiness for change. The Transtheoretical Model describes behavior change as a staged process that includes precontemplation, contemplation, preparation, action, and maintenance Prochaska and Velicer (1997); Raihan and Cogburn (2023). A client in precontemplation may minimize the problem or resist direct analysis, while a client in preparation may be more open to structured reflection, cognitive restructuring, or planning. Therefore, the same therapeutic move may be helpful in one stage but ineffective or even counterproductive in another. For example, introducing CBT too early may feel directive and increase resistance, whereas remaining only in MI after the client becomes ready may delay more structured therapeutic progress Arkowitz and Westra (2004); Marker and Norton (2018); Randall and McNeil (2017); Westra et al. (2009, 2016). In this thesis, resistance and readiness are treated as dynamic interaction states that shape whether an MI-style response, a CBT-style response, or a gradual transition between the two is more suitable.

These considerations motivate the need for LLM-based counseling systems that are sensitive not only to the current user message, but also to the likely direction of the dialogue. In therapeutic conversation, the goal is not simply to produce a locally appropriate response, but to choose a response that helps the client move toward greater engagement, openness, collaboration, and readiness. This is especially important in MI-CBT dialogue because different therapeutic moves may produce different near-future client reactions depending on the client's resistance, ambivalence, and stage of change. Therefore, the system needs a mechanism for comparing possible MI-style and CBT-style responses before committing to one response. Short-horizon lookahead provides one way to address this problem, because it allows multiple candidate responses and possible future client reactions to be considered before selecting the final response. Similar search-based ideas have been explored in large language model reasoning, where considering multiple candidate paths can improve decision-making in complex tasks Yao et al. (2023). This thesis adapts this general idea to simulated MI-CBT therapeutic dialogue by investigating whether reward-guided lookahead search can support better near-future therapeutic states.

To address this direction, this thesis proposes AIM-RGL, an agentic MI-CBT framework with reward-guided lookahead search for managing client resistance in simulated therapeutic conversations. The framework uses specialist therapist agents to generate candidate responses, including agents for open-ended questioning, reflection, affirmation, summarization, and CBT. A client simulator then produces possible future client reactions based on cognitive conceptualization and stage-of-change information, while a reward model evaluates candidate trajectories using therapeutic dimensions such as alliance, empathy, collaboration, CBT skill, MI quality, and client readiness. Instead of selecting only a locally appropriate response, AIM-RGL aims to select the response expected to support a better near-future therapeutic state. This thesis does not claim to diagnose, treat, replace professional therapists, or evaluate real clinical outcomes; rather, it investigates future-aware response selection in simulated AI-assisted counseling dialogue.

1.2 Statement of the Problem

Although LLM-based counseling systems have become increasingly capable of generating fluent and empathetic responses, many still treat therapeutic response generation mainly as an immediate-context decision. This creates a problem because therapeutic response quality depends not only on how appropriate a response appears in the present turn, but also on how it affects the client's next state. A response may sound supportive but still increase defensiveness, reduce engagement, or weaken the therapeutic alliance if it does not match the client's readiness for change Bordin (1979); Rogers (1957). Recent evaluations of LLM-based counseling systems also highlight the need to assess these systems using counseling-specific competencies, safety, and therapeutic quality rather than relying only on general language quality Hua et al. (2025); Nguyen et al. (2025). Therefore, the central problem addressed in this thesis is the difficulty of selecting therapeutically appropriate responses for resistant or ambivalent clients when response quality depends not only on current-turn suitability, but also on the client's likely next state.

This problem is especially important when an LLM-based system needs to coordinate MI and CBT responses for resistant or ambivalent clients. MI and CBT support different therapeutic needs: MI helps the client explore ambivalence and motivation, while CBT supports structured examination of thoughts and behavior change Beck (2020); Bischof et al. (2021); Chand et al. (2023); Miller and Rollnick (2013). However, the usefulness of each approach depends on when it is applied. If CBT is introduced before the client is ready, it may feel directive and increase resistance; if the system remains only in MI after the client becomes ready, it may miss opportunities for structured therapeutic progress Prochaska and Velicer (1997); Raihan and Cogburn (2023). Therefore, the problem is not simply choosing between MI and CBT, but selecting the therapeutic move that best fits both the client's current state and likely near-future response.

Prior work on LLM-based counseling has studied memory, stage awareness, strategy selection, and therapeutic response evaluation. For example, ChatThero uses a multi-session, stressor-aware language agent for recovery support, while Persona Drift explores adaptive flow in stage-aware CBT chatbots J. Wang et al. (2025); Yun, Kim, Kim, and Oh (2025). More closely related to this thesis, FLASH proposes a hybrid MI-CBT counseling framework that uses Flashbulb Memory to store high-impact client reactions to specific strategies and dynamically guide later intervention selection based on historical success and failure Anonymous (2026). However, these systems mainly adapt through memory, stage awareness, or previously observed interaction signals, and may still have limited ability to compare the immediate downstream effects of multiple candidate therapeutic moves before selecting a final response. This creates a gap between responses that fit the current turn and decisions that consider future dialogue states. This thesis addresses the gap by testing whether reward-guided lookahead search can compare multiple candidate MI-CBT responses and select the response expected to produce a better simulated client trajectory.

1.3 Objectives

The main objective of this thesis is to develop and evaluate AIM-RGL, an agentic MI-CBT framework that uses reward-guided lookahead search to improve therapeutic response selection for managing client resistance in simulated counseling dialogue.

- To build a multi-agent therapist architecture that generates candidate responses with specialist agents for open-ended questioning, reflection, affirmation, summarization, and CBT.
- To design a simulated client interaction setting grounded in cognitive conceptualization and stage-of-change information, so that client resistance, ambivalence, and readiness can be modeled during therapeutic dialogue.
- To train a reward model that estimates the therapeutic value of candidate responses based on dimensions such as therapeutic alliance, empathy, collaboration, CBT skill, MI quality, and client readiness.
- To implement reward-guided lookahead search that compares multiple candidate MI-CBT responses and selects the response expected to support a better near-future client trajectory.
- To examine how different lookahead horizons, including one-step, two-step, and three-step lookahead, affect the quality of MI-CBT therapeutic response selection.
- To evaluate whether AIM-RGL improves therapeutic skill quality, MI-CBT fidelity, therapeutic alliance, change talk, and sustain talk compared with selected baseline counseling systems.
- To analyze the effect of different reward model backbones and policy model backbones on the performance of reward-guided MI-CBT response generation.

1.4 Organization of the Study

This thesis is organized into five chapters. Chapter 1 introduces the research background, problem statement, objectives, and overall organization of the study. It explains the motivation for developing an LLM-based MI-CBT counseling framework that can manage client resistance through future-aware response selection. Chapter 2 reviews the literature related to therapeutic alliance, client resistance, readiness for change, Motivational Interviewing, Cognitive Behavioral Therapy, LLM-based counseling systems, client simulation, therapeutic evaluation, reward models, and lookahead search. This chapter establishes the clinical and technical foundations for AIM-RGL.

Chapter 3 presents the proposed methodology. It describes the overall AIM-RGL framework, including the multi-agent therapist response generation module, client simulation setting, reward-guided lookahead search mechanism, reward model training, and policy optimization process. Chapter 4 presents the experimental design used to evaluate AIM-RGL. It includes the research questions, hypotheses, independent variables, dependent variables, datasets, model backbones, baseline systems, lookahead configurations, evaluation metrics, and comparison procedures. Chapter 5 concludes the thesis by summarizing the main contributions, discussing the expected implications and limitations of the study, and outlining possible directions for future work.

CHAPTER 2
LITERATURE REVIEW

This chapter reviews the main theoretical and technical foundations related to this thesis. It first discusses therapeutic communication, client resistance, readiness for change, Motivational Interviewing, and Cognitive Behavioral Therapy as the clinical basis for MI-CBT response selection. It then reviews prior LLM-based counseling systems, including approaches that use memory, stage awareness, strategy selection, and therapeutic evaluation. Finally, the chapter connects these works to search-based and reward-guided methods, highlighting the gap that motivates AIM-RGL: the need for a system that can compare possible near-future client reactions before selecting a therapeutic response.

2.1 Clinical Foundations of MI-CBT Therapeutic Dialogue

Therapeutic dialogue differs from ordinary conversation because the therapist's response should support understanding, collaboration, and meaningful change. A counseling response is not evaluated only by relevance or fluency, but also by whether it strengthens the therapeutic relationship, respects client autonomy, and fits the client's current psychological state. This is especially important when the client shows resistance, ambivalence, or low readiness for change, because poorly timed responses may reduce engagement or increase defensiveness Beutler, Moleiro, and Talebi (2002); Patterson and Chamberlain (1994); Patterson and Forgatch (1985).

2.1.1 Therapeutic Alliance and Client-Centered Communication

Therapeutic alliance refers to the collaborative relationship between the therapist and the client. Bordin described the working alliance as involving agreement on goals, agreement on therapeutic tasks, and the emotional bond between therapist and client Bordin (1979). Rogers also emphasized empathy, genuineness, and unconditional positive regard as core conditions for therapeutic personality change Rogers (1957). These foundations suggest that therapeutic dialogue depends not only on the intervention itself, but also on how the intervention is communicated.

Client-centered communication is important in AI-assisted counseling because an LLM may generate a response that sounds supportive but still feels directive, assumptive, or poorly matched to the client's readiness. For example, a response may appear warm while still pushing the client toward action too quickly. In MI-CBT dialogue, this is important because structured CBT techniques may require sufficient trust and collaboration before they are introduced. Therefore, therapeutic alliance is treated as a key quality dimension in this thesis when evaluating whether a response supports engagement and readiness for further therapeutic work.

2.1.2 Client Resistance, Ambivalence, and Readiness for Change

Client resistance refers to reluctance, opposition, or disengagement during the therapeutic process. It may appear openly through arguing or rejecting suggestions, or more subtly through short answers, avoidance, topic changes, or passive agreement Patterson and Chamberlain (1994); Patterson and Forgatch (1985). Resistance has also been discussed as a process that may be influenced by therapist behavior, rather than only as a fixed client trait Beutler et al. (2002). In MI, resistance is often understood as a signal that the therapeutic approach may not fit the client's current readiness or concerns Miller and Rollnick (2013).

Ambivalence occurs when clients hold both reasons for and reasons against change. Motivational Interviewing addresses this tension by helping clients explore ambivalence and strengthen their own reasons for change Bischof et al. (2021); Miller and Rollnick (2013). Therapist responses can also influence whether clients produce more change talk or sustain talk during MI sessions Moyers et al. (2007). In this thesis, ambivalence is important because it creates moments where the system must decide whether the client needs further motivational support or is ready for structured CBT intervention.

Readiness for change explains why the same therapeutic response may help one client but hinder another. The Transtheoretical Model describes change as a staged process that includes precontemplation, contemplation, preparation, action, and maintenance Prochaska and Velicer (1997); Raihan and Cogburn (2023). A client in precontemplation may not yet view the issue as a problem, while a client in preparation may be more open to concrete strategies or behavioral steps. Therefore, therapeutic response selection should be sensitive to the client's current readiness rather than applying the same intervention style across all moments.

2.1.3 Motivational Interviewing, Cognitive Behavioral Therapy, and Their Integration

Motivational Interviewing is a counseling approach that supports motivation, autonomy, and readiness for change. It emphasizes collaboration, evocation, acceptance, and compassion, and commonly uses open-ended questions, affirmations, reflections, and summaries Bischof et al. (2021); Miller and Rollnick (2013). MI is especially useful when the client is ambivalent or resistant because it avoids direct confrontation and helps the client explore their own reasons for change. In this thesis, MI provides the foundation for response styles that reduce defensiveness and support readiness.

Cognitive Behavioral Therapy provides a more structured approach for identifying and modifying unhelpful thoughts, beliefs, and behaviors. CBT includes techniques such as identifying automatic thoughts, examining evidence, generating alternative interpretations, and planning behavioral change Beck (2020); Chand et al. (2023). These techniques are useful when the client is ready to reflect and take concrete steps, but they may feel directive if introduced too early. Therefore, CBT is most suitable when the client shows sufficient openness and the therapeutic relationship is stable.

The integration of MI and CBT has been explored in conventional mental health treatment, especially for clients who experience ambivalence or resistance during CBT. Prior work suggests that MI can support treatment engagement and prepare clients for more structured CBT work Arkowitz and Westra (2004); Westra et al. (2009, 2016). Reviews also indicate that MI may be useful as an adjunct to CBT, although the timing and manner of integration remain important Marker and Norton (2018); Randall and McNeil (2017).

Overall, these clinical foundations show that effective MI-CBT dialogue depends on resistance, ambivalence, readiness, and therapeutic alliance. MI and CBT offer complementary response styles, but their usefulness depends heavily on timing. This motivates AIM-RGL's focus on selecting responses not only by current-turn suitability, but also by their likely near-future effect on the simulated client's therapeutic state.

2.2 LLM-Based Counseling Systems and Therapeutic Evaluation

LLM-based counseling systems have recently been explored as tools for generating supportive, empathetic, and therapy-informed dialogue. However, counseling dialogue requires more than fluent response generation because the system must also consider therapeutic safety, client readiness, alliance, and the appropriateness of the intervention strategy Hua et al. (2025); Nguyen et al. (2025). For this thesis, prior systems are reviewed in terms of how they support CBT, MI, hybrid MI-CBT interaction, client simulation, and therapeutic evaluation.

2.2.1 LLM-Based CBT, MI, and Hybrid Counseling Agents

Existing LLM-based counseling systems support therapeutic dialogue in several ways, including CBT response generation, MI behavior-change support, counseling datasets, client simulation, and hybrid MI-CBT interaction. CBT-oriented works usually focus on structured therapeutic skills such as identifying thoughts, cognitive reframing, guided discovery, or generating multi-turn CBT-style counseling conversations. For example, CACTUS constructs multi-turn counseling dialogues grounded in CBT and client personas, while HealMe focuses on cognitive reframing through psychotherapy-informed dialogue Lee et al. (2024); Xiao et al. (2024). Psych8k and ChatCounselor further show the value of domain-specific counseling conversations for training mental health support models, as they use counseling conversations to improve counseling-related response quality Liu et al. (2023).

MI-based and behavior-change systems focus more on motivation, autonomy, readiness, and the client's movement toward change. CAMI is an MI-grounded counselor agent that uses state inference, topic exploration, action selection, and response generation to support behavior-change counseling Y. Yang, Achananuparp, Huang, Jiang, Kit, et al. (2025). CCSMI extends this direction by focusing on consistent client simulation for MI-based counseling, where the simulated client's mental state, motivation, beliefs, plans, and receptivity are tracked across the conversation Y. Yang, Achananuparp, Huang, Jiang, Lim, et al. (2025). These works are relevant to this thesis because they show how LLM-based counseling systems can represent client state and readiness during multi-turn interaction.

Other studies focus on constructing or reconstructing counseling dialogues for training and evaluation. SMILE converts single-turn mental health support data into multi-turn conversations, while CPsyCoun reconstructs Chinese psychological counseling dialogues from counseling reports and proposes an evaluation framework for multi-turn counseling Qiu, He, Zhang, Li, and Lan (2024); C. Zhang et al. (2024). PATIENT-Ψ is also relevant because it uses CBT-based cognitive models to simulate patients for training mental health professionals, making it closely related to cognitive conceptualization and simulated client design R. Wang et al. (2024). MAGNeT further extends synthetic counseling data generation by using coordinated specialist agents to generate multi-turn mental health counseling sessions Mandal, Chakraborty, and Gurevych (2025).

More recent work has also explored adaptive and hybrid counseling agents that are closer to the direction of this thesis. Persona Drift studies stage-aware adaptation in CBT chatbots by adjusting the dialogue flow when persona or stage drift is detected Yun et al. (2025). However, persona drift may act as an indirect signal because it reflects changes in the chatbot's generated persona rather than directly comparing possible future client reactions. ChatThero explores multi-session LLM-supported therapeutic support for behavior change by using memory-persistent and stressor-aware interaction across sessions J. Wang et al. (2025). Although this supports continuity and context-sensitive strategy use, it is less focused on explicitly anticipating the immediate downstream effects of a therapeutic move before the response is selected.

FLASH is especially close to this thesis because it combines MI and CBT and uses Flashbulb Memory to guide strategy selection for managing client resistance Anonymous (2026). Its memory-based approach allows the system to adapt from important previous client reactions and accumulated interaction history. However, this also means that adaptation depends strongly on observed history, and the system may have less support for comparing multiple possible near-future client reactions at the moment of response selection. Therefore, existing LLM-based counseling systems provide important foundations in CBT response generation, MI-based behavior change, client simulation, synthetic dialogue construction, and adaptive strategy selection, but they still leave room for future-aware response selection that evaluates possible next client states before choosing a therapeutic response. This gap motivates the reward-guided lookahead direction of AIM-RGL.

2.2.2 Client Simulation and Cognitive Conceptualization

Client simulation is useful for evaluating LLM-based counseling systems before involving real users, especially in sensitive mental health-related settings. In this thesis, simulated clients provide controlled interaction scenarios involving resistance, ambivalence, and different levels of readiness. Prior work has used LLMs to construct or simulate counseling interactions for training and evaluation, including PATIENT-Ψ, which uses CBT-based cognitive models to simulate patients for mental health professional training, and CCSMI, which focuses on consistent client simulation for MI-based counseling by tracking client mental states, motivation, beliefs, plans, and receptivity R. Wang et al. (2024); Y. Yang, Achananuparp, Huang, Jiang, Lim, et al. (2025). These studies show that client simulation can support controlled testing of counseling agents when direct evaluation with real clients is difficult or inappropriate.

Some recent counseling systems also use stage-of-change concepts from the Transtheoretical Model to represent client readiness during interaction. CAMI uses client state inference based on the Transtheoretical Model to support MI counseling, while CCSMI uses stages such as precontemplation, contemplation, and preparation to define and control simulated client states Prochaska and Velicer (1997); Raihan and Cogburn (2023); Y. Yang, Achananuparp, Huang, Jiang, Kit, et al. (2025); Y. Yang, Achananuparp, Huang, Jiang, Lim, et al. (2025). Persona Drift is also relevant because it studies stage-aware adaptation in CBT chatbots and adjusts the dialogue flow when persona or stage drift is detected Yun et al. (2025). This is relevant to this thesis because resistance and readiness are not fixed labels; they may change across turns depending on how the counselor responds.

Cognitive conceptualization can support client simulation by representing the client's situation, beliefs, emotions, automatic thoughts, coping patterns, and behavioral responses. In CBT, cognitive conceptualization is commonly used to understand how a person's thoughts, beliefs, emotions, and behaviors are connected within a specific problem context Beck (2020). For simulated therapeutic dialogue, this structure helps the client model respond more consistently across turns instead of producing isolated or random replies. Related work on synthetic and reconstructed counseling dialogue, such as CACTUS, SMILE, CPsyCoun, and MAGNeT, also shows the value of structured multi-turn dialogue construction for developing and evaluating mental health support systems Lee et al. (2024); Mandal et al. (2025); Qiu et al. (2024); C. Zhang et al. (2024). Therefore, cognitive conceptualization and client simulation provide an important foundation for evaluating AIM-RGL in controlled simulated MI-CBT conversations.

2.2.3 Evaluation Metrics for Therapeutic Dialogue

Therapeutic dialogue evaluation should consider counseling-specific qualities rather than only general language quality. In this thesis, evaluation focuses on five main dimensions: CBT competence, MI consistency, therapeutic alliance, change talk, and sustain talk. These dimensions are important because AIM-RGL is not only expected to generate fluent responses, but also to support appropriate therapeutic timing, collaboration, readiness, and movement toward change. The Cognitive Therapy Rating Scale (CTRS) provides a basis for evaluating CBT-related therapist competence, including structure, guided discovery, and cognitive intervention quality Beck Institute for Cognitive Behavior Therapy (2021); Young and Beck (1980). The Motivational Interviewing Treatment Integrity code (MITI) is used to assess MI fidelity, including empathy, autonomy support, and MI-consistent communication Moyers, Manuel, and Ernst (2015); Moyers, Rowell, Manuel, Ernst, and Houck (2016). The Working Alliance Inventory (WAI) provides a basis for evaluating therapeutic alliance through trust, collaboration, agreement on goals, and agreement on therapeutic tasks Hatcher and Gillaspy (2006); Horvath and Greenberg (1989).

Because therapeutic alliance is central to this thesis, it is represented through three components: goal agreement, approach or task agreement, and affective bond. Goal agreement refers to whether the client and counselor share an understanding of what therapy is trying to accomplish. Approach or task agreement refers to whether both sides appear aligned on the steps or activities used during the therapeutic process. Affective bond refers to the emotional connection between the client and counselor, including trust, confidence, appreciation, and mutual respect Bordin (1979); Hatcher and Gillaspy (2006); Horvath and Greenberg (1989). Table 2.1 summarizes the alliance-related questionnaire items used to evaluate these dimensions.

In addition to rating-based measures, this thesis also considers client language indicators from Motivational Interviewing. Change talk refers to client language that favors change, while sustain talk refers to client language that favors maintaining the current state Miller and Rollnick (2013); Miller and Rose (2009). These indicators are useful because they reflect whether the simulated client is moving toward readiness, motivation, and action, or remaining in resistance, ambivalence, or lower readiness. Therefore, change talk and sustain talk provide a simple way to observe the direction of the simulated client's

Table 2.1

Therapeutic alliance dimensions and questionnaire items.

Dimension	Question	No.
Goal	There is mutual understanding about what participants are trying to accomplish in therapy.	Q1
	The client and counselor are working on mutually agreed upon goals.	Q2
	The client and counselor have the same ideas about the client's real problems.	Q3
	The client and counselor have established a good understanding of the changes that would be good for the client.	Q4
Approach	There is agreement about the steps taken to help improve the client's situation.	Q5
	There is agreement about the usefulness of the current activity in therapy.	Q6
	There is agreement on what is important for the client to work on.	Q7
	The client believes that the way they are working with the problem is correct.	Q8
Affective Bond	There is a mutual liking between the client and counselor.	Q9
	The client feels confident in the counselor's ability to help the client.	Q10
	The client feels that the counselor appreciates him/her as a person.	Q11
	There is mutual trust between the client and counselor.	Q12

response after a therapeutic intervention.

The proportions of change talk and sustain talk can be represented as follows:

$$\text{Change Talk Ratio} = \frac{\text{Number of change talk utterances}}{\text{Total number of client utterances}} \tag{2.1}$$

$$\text{Sustain Talk Ratio} = \frac{\text{Number of sustain talk utterances}}{\text{Total number of client utterances}} \tag{2.2}$$

These ratios do not measure real clinical outcomes, but they are useful indicators for simulated therapeutic dialogue. A higher proportion of change talk may suggest greater readiness or movement toward change, while a higher proportion of sustain talk may suggest continued resistance, ambivalence, or attachment to the current state. Together, CTRS, MITI, WAI, change talk, and sustain talk allow this thesis to evaluate whether AIM-RGL improves therapeutic response quality in terms of CBT skill, MI consistency, alliance, readiness, and simulated client movement across the dialogue.

2.3 Reward-Guided Lookahead Search for MI-CBT Dialogue

Reward-guided lookahead search refers to choosing an action by considering not only its immediate suitability, but also its possible downstream effects. This idea is common in artificial intelligence, especially in sequential decision-making problems where a current decision changes the quality of later states Sutton and Barto (2018). In game playing, mathematical reasoning, instruction following, and text generation, systems often generate multiple candidate paths, evaluate them, and choose the option expected to lead to a better outcome. These methods are relevant to this thesis because therapeutic dialogue is also sequential: a therapeutic move that appears appropriate in the current turn may still lead to resistance, disengagement, or weaker alliance in the next turn.

2.3.1 Search-Based Reasoning and Lookahead Methods

Search-based methods have been successful in domains where decisions require planning over future possibilities. In game playing, AlphaGo combined deep neural networks with tree search to evaluate possible future moves before choosing an action, showing that learned evaluation and search can support strong sequential decision-making Silver et al. (2016). In language-model reasoning, Self-Consistency improves chain-of-thought reasoning by sampling multiple reasoning paths and selecting the most consistent answer rather than relying on a single greedy path X. Wang et al. (2022). Tree of Thoughts further extends this idea by allowing language models to explore, evaluate, and backtrack across multiple intermediate reasoning paths Yao et al. (2023). These works show that exploring alternatives before committing to a final output can improve performance in tasks that require planning or reasoning.

More recent work combines search with reward signals during LLM inference or self-training. Training Verifiers for math word problems generates multiple candidate solutions and chooses the one judged most likely to be correct by a verifier model Cobbe et al. (2021). ReST-MCTS* uses process-reward-guided Monte Carlo Tree Search to collect higher-quality reasoning traces and improve policy and reward models through self-training D. Zhang et al. (2024). Instead of relying only on final-answer correctness, these approaches evaluate intermediate steps or candidate paths. This suggests that lookahead-based planning is useful when the system must compare several possible paths before committing to one output.

In emotional support dialogue, lookahead planning has also been studied more directly. MultiESC improves multi-turn emotional support dialogue generation through lookahead strategy planning, where future user feedback is estimated after different support strategies so that the system can choose strategies expected to produce better long-term effects Cheng et al. (2022). This is relevant because emotional support dialogue, like therapeutic dialogue, requires the system to consider how a current therapeutic move may affect the user's later state.

Figure 2.1 illustrates the general idea of reward-guided lookahead search, inspired by search-based reasoning, process reward-guided tree search, and lookahead planning methods Cheng et al. (2022); Yao et al. (2023); D. Zhang et al. (2024).

2.3.2 Reward Models and Reward-Guided Candidate Evaluation

Reward models provide a way to convert human preferences, task success, or quality judgments into a score that can guide candidate evaluation or model training. In reinforcement learning from human preferences, Christiano et al. trained reward models from human comparisons of trajectory segments and used them to guide agents in Atari and simulated robotics tasks Christiano et al. (2017). In LLM alignment, InstructGPT trained a reward model from human rankings of model outputs and used reinforcement

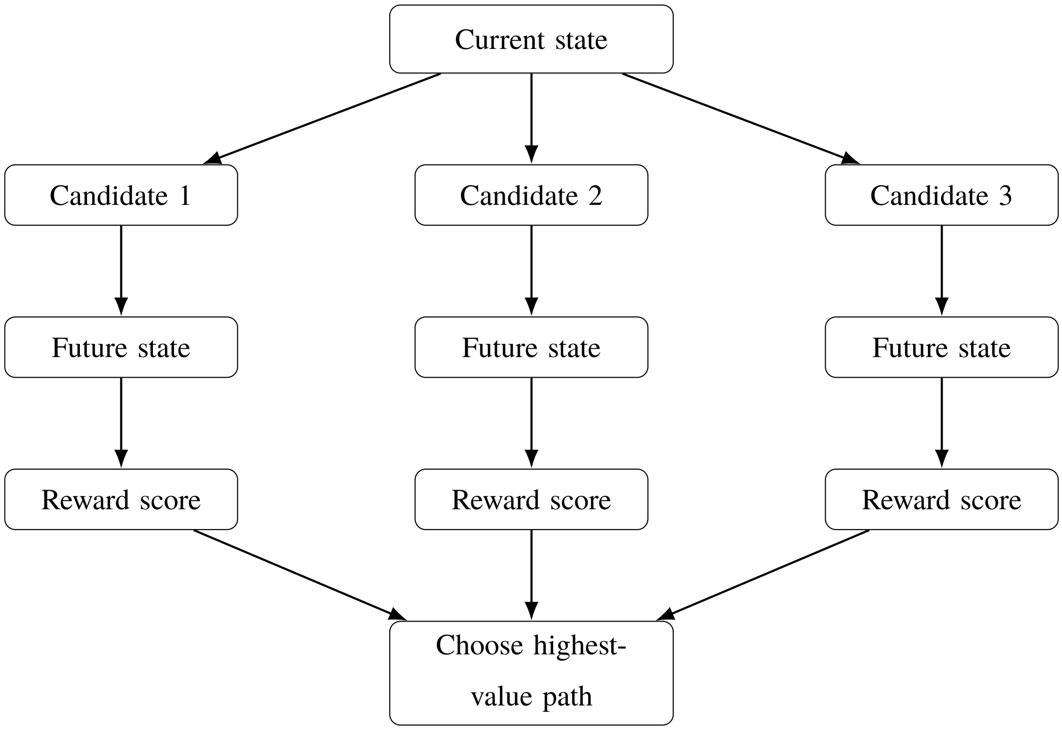

Figure 2.1

Generic reward-guided lookahead search for choosing among candidate action paths.

learning from human feedback to improve instruction-following behavior Ouyang et al. (2022). Direct Preference Optimization later showed that preference learning can also be optimized more directly without a separate reinforcement learning stage Rafailov et al. (2023). These works show that preference-based feedback can guide model behavior beyond simple next-token prediction.

Reward-guided candidate evaluation has also been used directly for choosing among generated outputs. Training Verifiers generates multiple candidate solutions for mathematical reasoning and chooses the solution judged most likely to be correct by a verifier model Cobbe et al. (2021). LLM-Blender uses PairRanker to compare candidate responses and GenFuser to combine high-ranked outputs, showing that ranking and candidate evaluation can improve generation quality across multiple LLMs Jiang, Ren, and Lin (2023). Process supervision further extends reward modeling by evaluating intermediate reasoning steps rather than only final answers, and has been shown to improve performance on challenging reasoning tasks Lightman et al. (2023). These methods support the idea that output quality can be improved by evaluating candidate outputs or intermediate paths before producing a final answer.

Dialogue-generation research has also explored feedback and reward signals for improving longer-horizon interaction quality. LSTDial uses both turn-level and dialogue-level measurement feedback during training, showing that dialogue agents can benefit from optimizing short-term output quality and longer-term conversation quality Ye, Zhao, Zhang, Zha, and Jiang (2024). In emotional support dialogue, RLFF-ESC uses multi-agent simulation to generate future dialogue trajectories, collect future-oriented rewards, train a future-oriented reward model, and improve an emotional support policy model T. Yang, Chen, and Wang (2025). These studies are relevant because supportive dialogue, like therapeutic dialogue, requires attention to how a current therapeutic move may affect later user states rather than only the immediate output quality.

Following reward-model and ranking-based candidate evaluation approaches, multiple candidate actions can be scored and the highest-valued option can be chosen Cobbe et al. (2021); Jiang et al. (2023); Ouyang et al. (2022):

$$a_t^* = \arg\max_{a_t^{(i)} \in \mathcal{A}_t} \hat{R}(H_t, a_t^{(i)}) \tag{2.3}$$

where H_t is the current context, $\mathcal{A}_t$ is the set of candidate actions, $a_t^{(i)}$ is the i-th candidate action, $\hat{R}$ is the estimated reward function, and a_t^* is the chosen action. This formulation represents the general idea of choosing the candidate action with the highest estimated value.

When future outcomes are considered, the value of a candidate trajectory can be represented using the standard discounted return formulation from reinforcement learning Sutton and Barto (2018):

$$G_t^{(i)} = \sum_{k=0}^{d-1} \gamma^k \hat{R}_{t+k}^{(i)} \tag{2.4}$$

where $G_t^{(i)}$ is the estimated value of candidate trajectory i, d is the lookahead depth, γ is the discount factor, and $\hat{R}_{t+k}^{(i)}$ is the predicted reward at future step k. In this thesis, these equations are used only as generic formulations for reward-guided choice and future-aware trajectory evaluation. The detailed AIM-RGL formulation is presented later in the methodology chapter.

2.3.3 Application to MI-CBT Strategy Coordination

The reviewed methods show that search-based reasoning, lookahead planning, reward modeling, and candidate ranking can improve decision-making in several domains. In games, search helps evaluate future consequences before choosing a move. In reasoning tasks, lookahead allows models to compare multiple paths instead of committing to the first generated solution. In alignment, dialogue generation, and emotional support systems, reward models and feedback signals help choose or train outputs that better match task goals, human preferences, or longer-horizon interaction quality.

In this thesis, these methods are not treated as entirely new technical ideas. Instead, they are used as foundations for addressing the strategy-coordination problem in simulated MI-CBT therapeutic dialogue. Prior LLM-based counseling systems have explored CBT-style response generation, MI-based behavior-change support, client simulation, stage-aware adaptation, and memory-based strategy planning. These systems provide important foundations, but the MI-CBT setting creates a specific timing challenge: the system must decide whether the client needs motivational support, structured cognitive intervention, or a gradual transition between the two.

AIM-RGL is therefore positioned as an application of future-aware, reward-guided lookahead search to simulated MI-CBT counseling dialogue. Rather than relying only on the current turn, the framework uses lookahead to compare candidate therapeutic moves according to their likely near-future effects on the simulated client. This positioning connects prior work on search and reward-guided candidate evaluation with the clinical challenge of coordinating MI and CBT at the right moment. The detailed AIM-RGL framework and implementation are presented in Chapter 3.

CHAPTER 3
METHODOLOGY

This chapter presents the methodology used to develop AIM-RGL, an agentic MI-CBT framework with reward-guided lookahead search for therapeutic response generation. The chapter describes how the proposed system generates multiple candidate therapist responses, evaluates their possible near-future effects, and selects responses expected to support better therapeutic dialogue outcomes. The methodology is organized around the main AIM-RGL pipeline: multi-agent therapist response generation, reward-guided lookahead search, reward model training, and policy optimization. Client simulation and scenario construction are described as the supporting environment used to generate controlled therapist-client interactions. The study is limited to simulated counseling dialogue and does not aim to diagnose, treat, replace professional therapists, or measure real clinical outcomes.

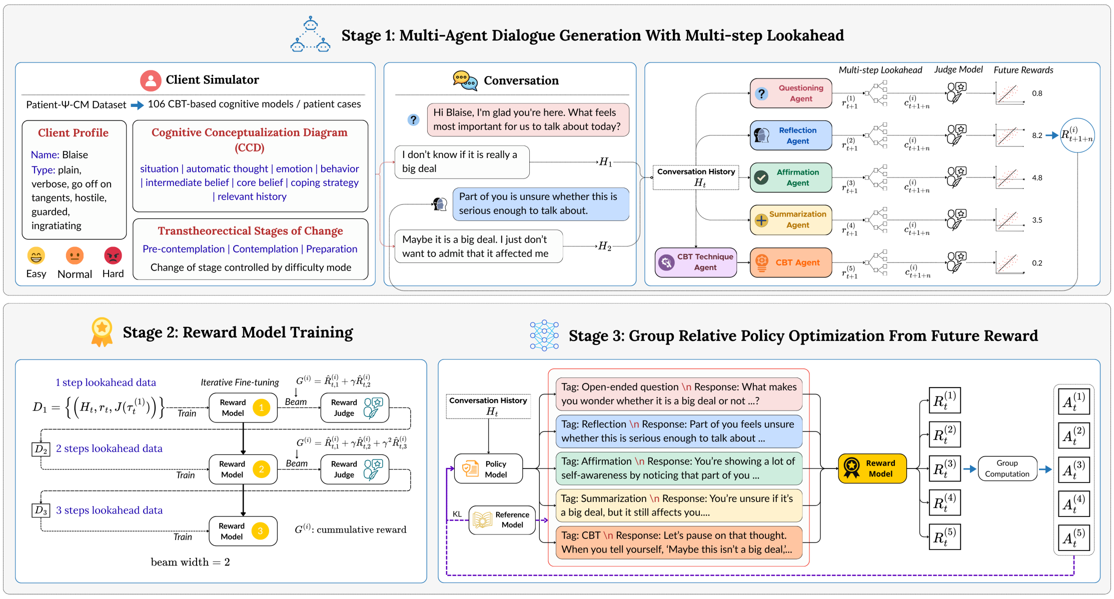

Figure 3.1

Overall AIM-RGL workflow, including multi-agent dialogue generation with lookahead, reward model training, and policy optimization from future reward.

3.1 Research Design and AIM-RGL Framework Overview

This study adopts a simulation-based system development methodology to design and evaluate AIM-RGL, an agentic MI-CBT framework with reward-guided lookahead search. The main purpose of the methodology is to construct a controlled dialogue framework in which multiple therapeutic responses can be generated, expanded through short-horizon future simulation, evaluated using therapeutic reward signals, and used to guide final response generation. The study focuses on simulated therapist-client dialogue and does not aim to diagnose, treat, replace professional therapists, or measure real clinical treatment outcomes.

AIM-RGL is designed for therapeutic conversations in which the client may show resistance, ambivalence, or changing readiness for structured intervention. In this setting, a response that appears appropriate in the current turn may not necessarily lead to a better next client state. Therefore, the framework treats MI-CBT response generation as a future-aware decision problem. Instead of generating only one therapist response directly, AIM-RGL first generates multiple candidate responses from specialist therapist agents, estimates possible future client reactions, and evaluates the resulting trajectories using therapeutic reward scores.

The overall framework consists of four main components. First, specialist therapist agents generate candidate responses based on different therapeutic skills, including open-ended questioning, reflection, affirmation, summarization, and CBT intervention. Second, reward-guided lookahead search expands these candidates by considering possible near-future dialogue outcomes. Third, a reward model estimates the therapeutic value of candidate responses or trajectories using dimensions such as alliance, empathy, collaboration, CBT skill, MI quality, and client readiness. Fourth, the learned reward signal is used to support policy optimization so that the response generation model can produce therapeutically stronger responses.

The methodology is organized according to this pipeline. Section 3.2 describes the multi-agent therapist response generation process. Section 3.3 explains the reward-guided lookahead search mechanism. Section 3.4 presents reward model training and policy optimization. Section 3.5 describes the simulation environment and client scenario construction used to support the framework. Finally, Section 3.6 discusses the implementation scope and ethical considerations of the study.

3.2 Multi-Agent Therapist Response Generation

AIM-RGL uses a multi-agent therapist architecture to generate candidate responses before the final therapeutic response is selected. The purpose of this stage is not to decide the best response immediately, but to produce a diverse set of therapeutically meaningful options that can later be evaluated through reward-guided lookahead search. Each therapist agent represents a different therapeutic skill or intervention style. This design makes the response generation process more explicit because the framework can compare different MI-style and CBT-style responses instead of relying on a single model to implicitly decide the strategy.

At each therapist turn, the system receives the current conversation history as input. The conversation history includes previous therapist and client utterances, the current client concern, and any available information about the client's readiness or resistance. Based on this context, the therapist agents generate multiple candidate responses. Each candidate response is tagged with the agent type that produced it, such as open-ended question, reflection, affirmation, summarization, or CBT intervention. These tagged responses form the candidate response pool for the next stage of the framework.

The MI-specific agents are based on the OARS communication skills commonly used in Motivational Interviewing: open-ended questions, affirmations, reflections, and summaries. The open-ended questioning agent generates questions that invite the client to explore thoughts, feelings, goals, and ambivalence in greater depth. The reflection agent mirrors the client's meaning, emotion, or ambivalence in order to show understanding and reduce defensiveness. The affirmation agent highlights the client's strengths, effort, values, or persistence in a specific and genuine way. The summarization agent gathers important themes from the conversation and presents them in an organized way to support transition, clarity, or continued exploration.

In addition to MI-specific agents, AIM-RGL includes CBT-specific response generation. The CBT component is divided into two steps. First, a CBT technique selector identifies a suitable CBT technique and plan step for the client's current concern. Possible techniques include identifying automatic thoughts, evidence-based questioning, alternative perspective taking, decatastrophizing, problem solving, and behavioral experiments. Second, the CBT response agent converts the selected technique and plan step into a brief and natural therapist response. The generated response is written in conversational language and does not need to explicitly name the CBT technique.

The role of the multi-agent stage is to create a balanced response pool that includes both supportive MI-style responses and structured CBT-style responses. This is important because resistant or ambivalent clients may not always be ready for direct cognitive restructuring or action planning. In some cases, a reflective or open-ended MI response may better preserve alliance and reduce resistance. In other cases, when the client shows readiness, a CBT response may help the client examine thoughts or take a concrete step. By generating these alternatives separately, AIM-RGL can later evaluate which response is more likely to support a better near-future therapeutic state.

Table 3.1 summarizes the main therapist agents used in AIM-RGL.

Table 3.1

Therapist agents used for candidate response generation.

Agent	**Function**
Open-ended Questioning Agent	Generates questions that encourage the client to explore thoughts, feelings, goals, and ambivalence.
Reflection Agent	Reflects the client's meaning, emotion, or ambivalence to show understanding and reduce defensiveness.
Affirmation Agent	Highlights the client's strengths, effort, values, or persistence to support confidence and engagement.
Summarization Agent	Organizes key themes, emotions, and concerns from the conversation to support clarity and transition.
CBT Technique Selector	Selects a suitable CBT technique and plan step based on the client's current concern and readiness.
CBT Response Agent	Generates a natural therapist response using the selected CBT technique and plan step.

The output of this stage is a set of tagged candidate therapist responses. These candidates are not treated as final responses. Instead, they are passed to the reward-guided lookahead search stage, where each response is expanded through simulated future client reactions and evaluated according to its expected therapeutic value.

3.3 Reward-Guided Lookahead Search

After the therapist agents generate candidate responses, AIM-RGL applies reward-guided lookahead search to select the response with the highest expected therapeutic value. The purpose of this stage is to avoid selecting a response only because it appears appropriate in the current turn. Instead, the framework estimates how each candidate response may influence the simulated client's near-future state, including engagement, resistance, readiness, and therapeutic alliance.

At therapist turn t, the conversation history is denoted as H_t. The multi-agent therapist module generates a set of candidate responses:

$$A_t = \{r_t^{(1)}, r_t^{(2)}, \dots, r_t^{(N)}\}$$

where $r_t^{(i)}$ is the candidate response generated by therapist agent i, and N is the number of candidate responses. In this thesis, the main candidate responses are generated by the open-ended questioning, reflection, affirmation, summarization, and CBT agents. Figure 3.2 illustrates this one-step reward-guided lookahead process. Starting from the current conversation history, AIM-RGL generates multiple therapist candidate responses, simulates one possible client response for each candidate, assigns a reward score to each resulting trajectory, and selects the response with the highest reward.

For each candidate response, the client simulator generates a possible future client response. This produces a one-step future trajectory for each candidate:

$$\tau_t^{(i,1)} = (H_t, r_t^{(i)}, c_{t+1}^{(i)})$$

where $c_{t+1}^{(i)}$ is the simulated client response after candidate response $r_t^{(i)}$. The resulting trajectory is evaluated by a reward judge or reward model:

$$R_t^{(i)} = J(H_t, r_t^{(i)}, c_{t+1}^{(i)})$$

where J represents the reward judge, and $R_t^{(i)}$ is the reward score assigned to candidate i. The reward score estimates the therapeutic value of the candidate response and its im-

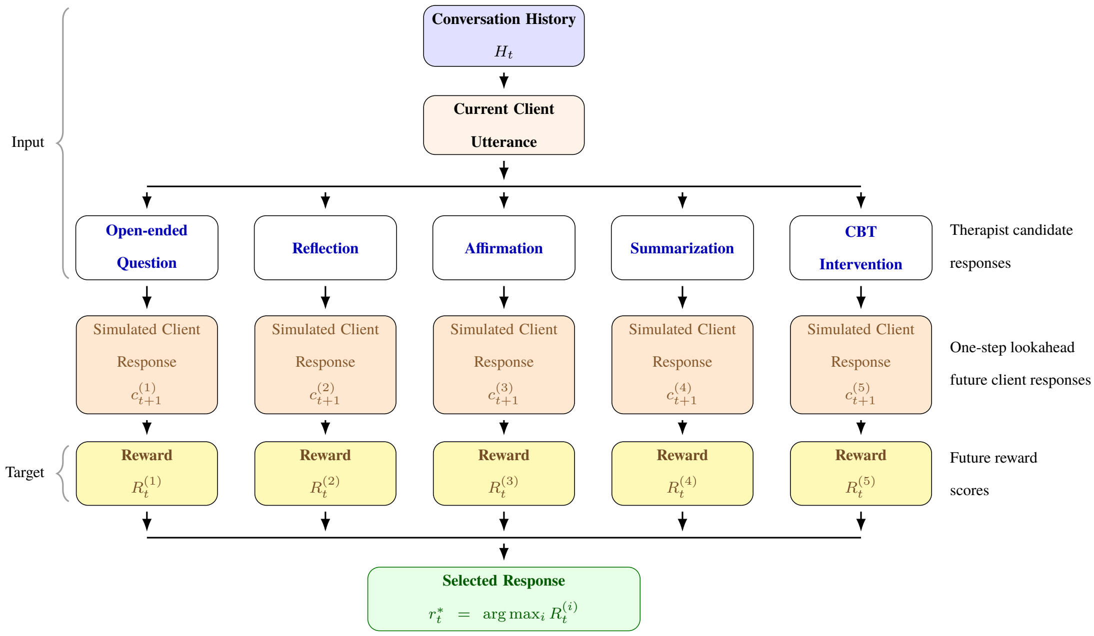

Figure 3.2

One-step reward-guided lookahead search for evaluating therapist candidate responses.

mediate future effect. It is based on therapeutic dimensions such as therapeutic alliance, empathy, collaboration, CBT skill, MI quality, and client readiness.

The selected therapist response is the candidate with the highest estimated reward:

$$r_t^* = \arg\max_{r_t^{(i)} \in A_t} R_t^{(i)}$$

This one-step formulation allows AIM-RGL to compare how different MI-CBT responses may affect the client immediately after the therapist turn. For example, a CBT response may appear useful in the current turn, but if the simulated client reacts with stronger resistance or lower readiness, its reward score may be lower than an MI-style reflection or open-ended question.

AIM-RGL also extends this process to multi-step lookahead. Instead of evaluating only the next client response, the framework estimates the cumulative therapeutic value of a response trajectory over a lookahead depth d. The cumulative value of trajectory i is computed as:

$$G_t^{(i)} = \sum_{k=1}^{d} \gamma^{k-1} \hat{R}_{t,k}^{(i)}$$

where $G_t^{(i)}$ is the cumulative reward of trajectory i, d is the lookahead depth, γ is the discount factor, and $\hat{R}_{t,k}^{(i)}$ is the predicted reward at lookahead step k. When $\gamma = 1$, all future steps are treated equally. When $\gamma < 1$, earlier rewards are given greater importance than later rewards.

For deeper lookahead, fully expanding all possible branches becomes computationally expensive because each therapist candidate can lead to multiple future client states and additional therapist responses. To reduce this cost, AIM-RGL uses beam-based pruning. After the one-step reward model RM_1 is trained, it is used to estimate the reward of first-step candidate responses. Only the top-ranked candidates are retained as beam branches and expanded further to construct the two-step lookahead dataset D_2. The same procedure is later applied using RM_2 to support three-step lookahead data generation. In this way, the framework avoids expanding the entire search tree while still preserving the most promising therapeutic trajectories for further evaluation.

The final selected response is therefore not simply the response with the best immediate wording. It is the response expected to produce the most beneficial near-future therapeutic trajectory. In the MI-CBT setting, this is important because the system must decide whether the client needs further motivational support, a structured CBT intervention, or a gradual transition between MI and CBT.

3.4 Reward Model Training and Policy Optimization

After candidate trajectories are generated through reward-guided lookahead search, AIM-RGL trains a reward model to estimate the therapeutic value of therapist responses without requiring the reward judge to score every candidate during later stages. The reward model is trained as a regression model that predicts a scalar reward score from the conversation history and a candidate therapist response. This allows the framework to approximate therapeutic reward more efficiently during multi-step lookahead and policy optimization.

Reward model training is carried out through an iterative fine-tuning process across increasing lookahead depths. The reward model is first trained on one-step lookahead data, where each candidate therapist response is evaluated after one simulated client response. After this first-stage reward model is trained, it is used to guide beam-based generation of two-step lookahead data by retaining only the most promising candidate branches. The same process is then repeated for three-step lookahead, allowing the reward model to gradually learn from deeper therapeutic trajectories while reducing the cost of expanding all possible dialogue branches. Figure 3.3 summarizes this iterative reward model training process.

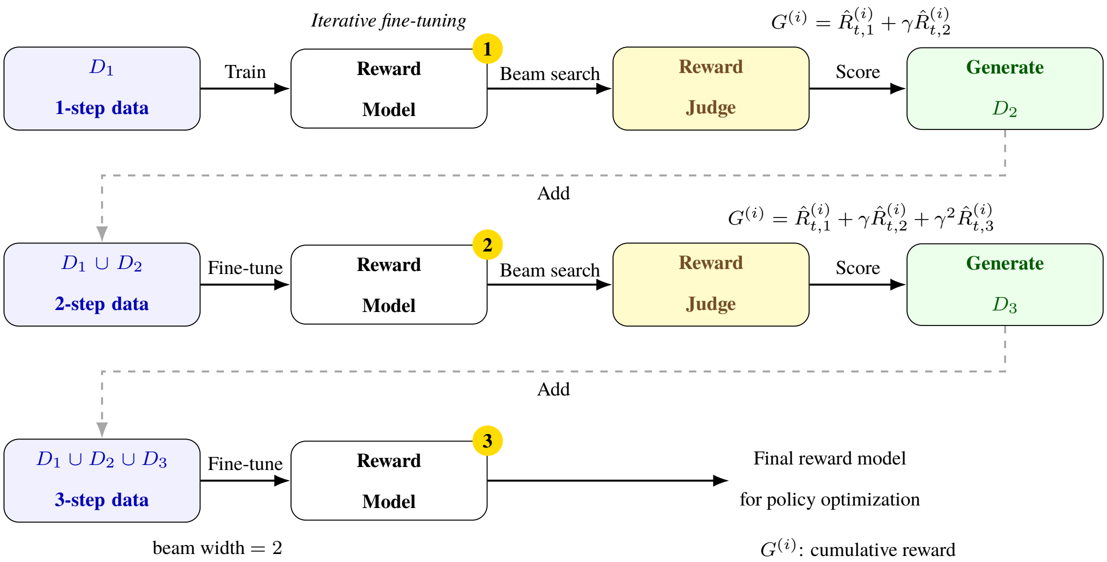

Figure 3.3

Iterative reward model fine-tuning with beam-guided generation of multi-step lookahead data.

The reward target is first produced by a reward judge. For each candidate trajectory, the reward judge evaluates the response according to multiple therapeutic dimensions, including therapeutic alliance, understanding and empathy, collaboration and structure, CBT skill and strategy, and MI quality and readiness. The overall reward score is computed as the average of these dimension scores:

$$R = \frac{S_{\text{alliance}} + S_{\text{empathy}} + S_{\text{collaboration}} + S_{\text{CBT}} + S_{\text{MI}}}{5}$$

where each S represents the score assigned to one therapeutic dimension. This scalar reward score is used as the training target for the reward model.

Let f_θ denote the trained reward model with parameters θ. Given the conversation history H_t and a candidate therapist response $r_t^{(i)}$, the reward model predicts:

$$\hat{R}_t^{(i)} = f_\theta(H_t, r_t^{(i)})$$

where $\hat{R}_t^{(i)}$ is the predicted therapeutic reward for candidate response i. The model is trained to minimize the difference between the predicted reward $\hat{R}_t^{(i)}$ and the reward target $R_t^{(i)}$ assigned by the reward judge. The training objective is formulated using mean squared error:

$$\mathcal{L}_{RM} = \frac{1}{M}\sum_{m=1}^{M}\left(\hat{R}_m - R_m\right)^2$$

where M is the number of training samples, $\hat{R}_m$ is the reward predicted by the reward model, and R_m is the reward target assigned by the reward judge.

Reward model training is performed iteratively across different lookahead depths. In the one-step stage, the system constructs the dataset D_1 by generating candidate therapist responses, simulating one future client response for each candidate, and assigning reward scores to the resulting one-step trajectories. The first reward model RM_1 is trained using this one-step lookahead dataset:

$$RM_1 = Train(M_0, D_1)$$

where M_0 is the pretrained base model used as the initial reward model backbone.

After RM_1 is trained, it is used to support two-step lookahead data generation. Instead of expanding all candidate branches, RM_1 scores the first-step candidate responses and retains only the top-ranked candidates as beam branches. These retained branches are then expanded further to generate two-step trajectories. The resulting dataset is denoted as D_2. The second reward model RM_2 is then fine-tuned using both one-step and two-step lookahead data:

$$D_2 = Generate(RM_1, d = 2)$$

$$RM_2 = FineTune(RM_1, D_1 \cup D_2)$$

The same procedure is repeated for three-step lookahead. The trained RM_2 is used to guide branch selection and generate the three-step lookahead dataset D_3. The third reward model RM_3 is then fine-tuned using all available lookahead datasets:

$$D_3 = Generate(RM_2, d = 3)$$

$$RM_3 = FineTune(RM_2, D_1 \cup D_2 \cup D_3)$$

This iterative training process allows the reward model to gradually learn from deeper future trajectories while reducing the computational cost of exhaustive search. The beam-based procedure keeps the most promising branches for expansion, making multi-step lookahead more practical.

After reward model training, the learned reward signal is used to optimize the therapist response policy. For each conversation history H_t, the policy model samples a group of candidate therapist responses:

$$r_t^{(1)}, r_t^{(2)}, \ldots, r_t^{(G)} \sim \pi_\phi(\cdot \mid H_t)$$

where π_ϕ is the policy model and G is the number of sampled responses in the group. Each sampled response is scored by the trained reward model:

$$R_i = RM(H_t, r_t^{(i)})$$

The rewards are then normalized within the group to compute a relative advantage for each candidate response:

$$A_i = \frac{R_i - \mu_R}{\sigma_R + \epsilon}$$

where μ_R and σ_R are the mean and standard deviation of the reward scores within the group, and ϵ is a small constant used for numerical stability. Responses with higher relative reward receive positive advantages, while lower-scoring responses receive lower or negative advantages.

3.4.1 GRPO

After the reward model is trained, it is used to guide policy optimization. For each conversation history, the policy model samples a group of candidate therapist responses. The trained reward model scores each response, and the scores are normalized within the group to compute relative advantages. These advantages are then used to update the policy model while a KL penalty constrains the updated policy from drifting too far from the reference model. Figure 3.4 illustrates this GRPO-based policy optimization process.

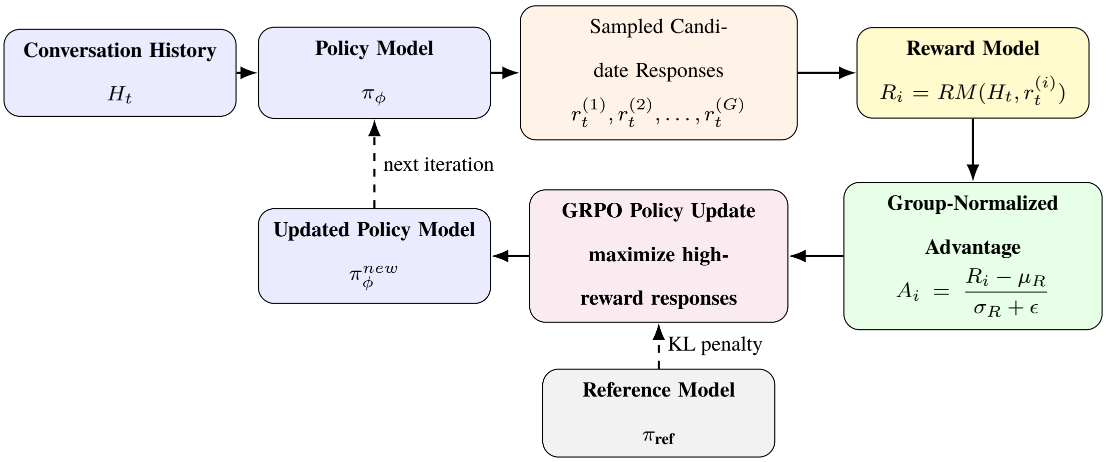

Figure 3.4

GRPO-based policy optimization using rewards predicted by the trained reward model.

The policy model is optimized using a group-relative objective. The purpose is to increase the likelihood of responses with higher predicted therapeutic reward while constraining the updated policy from drifting too far from the reference model. This is represented as:

$$\mathcal{L}_{GRPO}(\phi) = -\frac{1}{G}\sum_{i=1}^{G} \min\left(\rho_i A_i, \text{clip}(\rho_i, 1-\epsilon, 1+\epsilon)A_i\right) + \beta D_{KL}(\pi_\phi \| \pi_{\text{ref}})$$

where

$$\rho_i = \frac{\pi_\phi(r_t^{(i)} \mid H_t)}{\pi_{\text{old}}(r_t^{(i)} \mid H_t)}$$

ρ_i is the probability ratio between the updated policy and the old policy, A_i is the group-normalized advantage, β controls the strength of the KL penalty, and π_{ref} is the reference policy model.

Through this process, AIM-RGL uses the trained reward model not only for response selection, but also for improving the response generation policy. The policy optimization stage encourages the model to generate therapist responses that are more likely to support therapeutic alliance, empathy, collaboration, MI-CBT fidelity, and client readiness in simulated dialogue.

CHAPTER 4
EXPERIMENT

This chapter presents the experimental design used to evaluate AIM-RGL in simulated MI-CBT counseling dialogue. The purpose of the experiment is to examine whether reward-guided lookahead search can improve therapeutic response quality when compared with baseline response generation methods. The chapter describes the experimental conditions, including the simulated client setup, compared systems, lookahead configurations, reward model backbones, and policy model backbones. It also explains the evaluation protocol and metrics used to assess therapeutic skill quality, MI-CBT fidelity, therapeutic alliance, change talk, sustain talk, and reward model performance. Since the implementation and evaluation are still in progress, the final section of this chapter is reserved for reporting the experimental results after they are obtained.

4.1 Experimental Design and Conditions

4.1.1 Experiment Overview

This experiment is designed to evaluate AIM-RGL in simulated MI-CBT counseling dialogue. The main purpose is to examine whether reward-guided lookahead search can improve therapeutic response generation when the simulated client shows resistance, ambivalence, or changing readiness for structured intervention. In this setting, a response is not evaluated only by whether it appears appropriate in the current turn, but also by whether it is expected to support a better near-future therapeutic trajectory.

The experiment compares AIM-RGL with selected baseline therapist systems under the same simulated client scenarios. Each system receives the same client profile, cognitive conceptualization information, difficulty setting, and dialogue context. The resulting therapist-client conversations are evaluated using counseling-specific metrics, including therapeutic skill quality, MI-CBT fidelity, therapeutic alliance, change talk, and sustain talk. The experiment also examines the effects of different lookahead horizons, reward model backbones, and policy model backbones.

The experiment is guided by the following research questions:

- **RQ1:** Does reward-guided lookahead search improve therapeutic skill quality and MI-CBT fidelity compared with baseline systems?

- **RQ2:** Does reward-guided lookahead search strengthen therapeutic alliance compared with baseline systems?
- **RQ3:** Does reward-guided lookahead search promote earlier increases in change talk and reductions in sustain talk compared with baseline systems?
- **RQ4:** How do different lookahead horizons, including one-step, two-step, and three-step lookahead, affect reward-guided MI-CBT response quality?
- **RQ5:** How does the choice of policy model backbone affect the quality of MI-CBT therapeutic responses?

Based on these research questions, the experiment tests the following hypotheses:

- **H1:** Reward-guided lookahead search will produce higher therapeutic skill quality and stronger MI-CBT fidelity than baseline methods.
- **H2:** Reward-guided lookahead search will lead to stronger therapeutic alliance compared with baseline methods.
- **H3:** Reward-guided lookahead search will promote earlier increases in change talk and earlier reductions in sustain talk compared with baseline methods.
- **H4:** Longer lookahead horizons will improve MI-CBT response quality by considering future client reactions and therapeutic trajectory.
- **H5:** The choice of policy model backbone will affect the quality of MI-CBT therapeutic responses in simulated resistant-client conversations.

This experiment is limited to simulated therapist-client dialogue. It does not involve real patients or human clinical participants, and it does not measure real clinical treatment outcomes.

4.1.2 Experimental Conditions

The experiment includes four main experimental factors: therapist system, lookahead horizon, reward model backbone, and policy model backbone. The therapist system comparison is used to evaluate whether AIM-RGL improves therapeutic response quality compared with selected baseline systems. The lookahead horizon comparison is used to examine whether considering more future client reactions improves response selection. The reward model backbone comparison is used to evaluate reward prediction performance, while the policy model backbone comparison is used to examine whether policy optimization results depend on the underlying response generation model.

The therapist systems compared in this experiment are CBT, CAMEL, FLASH, and

AIM-RGL. The CBT baseline represents a structured cognitive-behavioral response generation approach. It focuses on CBT-style skills such as identifying thoughts, examining evidence, and supporting structured reflection or action. CAMEL is included as a baseline counseling agent for comparison Lee et al. (2024). FLASH is included as a memory-guided MI-CBT baseline that uses previous client interactions to guide therapeutic strategy selection Anonymous (2026). AIM-RGL is the proposed framework, which uses specialist therapist agents, client simulation, reward-guided lookahead search, reward model training, and policy optimization.

For AIM-RGL, the lookahead horizon is varied using one-step, two-step, and three-step lookahead settings. These are not treated as separate system names, but as different configurations of the same reward-guided lookahead mechanism. In one-step lookahead, each candidate therapist response is followed by one simulated client response before scoring. In two-step lookahead, the system expands the trajectory further to estimate a longer short-term therapeutic effect. In three-step lookahead, the system examines whether deeper trajectory evaluation provides additional improvement in response quality.

Table 4.1

Experimental conditions.

Condition Type	**Experimental Settings**
Therapist system	CBT, CAMEL, FLASH, AIM-RGL
Lookahead horizon	One-step lookahead, two-step lookahead, three-step lookahead
Reward model backbone	DeBERTa-v3-large, ModernBERT-large, RoBERTa-large
Policy model backbone	LLaMA-3.1-8B-Instruct, Qwen-2.5-7B-Instruct-1M

The reward model backbone comparison includes DeBERTa-v3-large, ModernBERT-large, and RoBERTa-large. These models are used to examine how different encoder-based architectures affect scalar therapeutic reward prediction. The policy model backbone comparison includes LLaMA-3.1-8B-Instruct and Qwen-2.5-7B-Instruct-1M. These models are used to examine how the choice of response generation backbone affects optimized MI-CBT therapeutic response quality.

4.1.3 Dataset and Client Simulation Setup

The experiment uses simulated client scenarios based on the Patient-Ψ-CM dataset R. Wang et al. (2024). The dataset contains CBT-based cognitive models and patient cases. Each case includes cognitive conceptualization information such as the client's situation, automatic thoughts, emotions, behaviors, coping strategies, relevant history, intermediate beliefs, and core beliefs. These elements are used to ground the simulated client's responses and maintain consistency across multi-turn dialogue.

The client simulator generates first-person client responses based on the current conversation history, the client's cognitive conceptualization profile, and the client's current stage of change. The stage of change is represented using three states: precontemplation, contemplation, and preparation. Precontemplation represents a resistant client who may minimize the problem, reject feedback, or show little interest in change. Contemplation represents an ambivalent client who recognizes both reasons for and against change. Preparation represents a client who is more open to structured reflection, planning, or CBT-style intervention.

Client difficulty is represented using easy, normal, and hard settings. Difficulty controls how frequently the client's openness level and stage of change are re-evaluated during the dialogue. Easy clients are expected to become more open more quickly, normal clients change more gradually, and hard clients remain resistant for longer. The openness judge periodically re-evaluates the simulated client's openness level and stage of change based on the recent conversation history. For easy clients, openness is re-evaluated every two turns. For normal clients, it is re-evaluated every four turns. For hard clients, it is re-evaluated every six turns. This interval-based design allows the simulation to represent different levels of resistance persistence and readiness development.

GPT-4o-mini is used for therapist-side response generation and client simulation. In AIM-RGL, the therapist-side generation module includes specialist agents for open-ended questioning, reflection, affirmation, summarization, and CBT intervention. These agents generate candidate responses before the reward-guided lookahead search stage. The reward model is trained to predict scalar therapeutic reward scores for candidate responses or trajectories. GPT-4o is used as a reward judge and evaluator for therapeutic quality dimensions, including therapeutic alliance, MI fidelity, and CBT competence.

The same model, dataset, and client simulation setup are applied across compared systems to support controlled comparison.

4.1.4 Reward Model and Policy Training Configuration

This subsection describes the training configuration used for the reward model and the policy optimization stage. The reward model is trained before policy optimization because it provides the scalar therapeutic reward signal used to score candidate therapist responses during reward-guided lookahead search and GRPO. The methodological details of reward model training and GRPO are already described in Chapter 3; therefore, this subsection focuses only on the experimental configuration, dataset size, model backbones, and training setup.

For reward model training, 10 simulated client cases are selected from the Patient-Ψ-CM dataset. Each client case is evaluated under three difficulty modes: easy, normal, and hard. For each client and difficulty mode, 29 scored therapist turns are used. In the one-step lookahead setting, five candidate therapist responses are generated for each scored turn, corresponding to the main therapist agents. Therefore, the one-step reward dataset contains $10 \times 3 \times 29 \times 5 = 4{,}350$ samples.

For deeper lookahead, the reward dataset is generated using beam-guided trajectory expansion. In the two-step lookahead setting, four judged trajectories are stored for each scored turn, producing $10 \times 3 \times 29 \times 4 = 3{,}480$ samples. The three-step lookahead setting follows the same sample structure and also produces $10 \times 3 \times 29 \times 4 = 3{,}480$ samples. In total, the planned reward model training dataset contains 11,310 samples across one-step, two-step, and three-step lookahead data.

The reward model is trained as a regression model. Its objective is to predict a scalar therapeutic reward score for each candidate response or lookahead trajectory. The reward targets are generated by the reward judge based on therapeutic dimensions such as therapeutic alliance, understanding and empathy, collaboration and structure, CBT skill and strategy, and MI quality and readiness. The dataset is divided into training, validation, and test sets using a 70%, 15%, and 15% split. Mean squared error loss is used for reward model training because the reward target is represented as a continuous scalar score.

After the reward model is trained, it is used to support policy optimization through GRPO. In this stage, the policy model generates a group of candidate therapist responses for the same conversation context. The trained reward model scores each candidate response, and the group-level reward distribution is used to compute relative advantages. The policy model is then optimized to increase the likelihood of higher-reward therapeutic responses while remaining close to the reference model through a KL penalty.

The GRPO stage compares two policy model backbones: LLaMA-3.1-8B-Instruct and Qwen-2.5-7B-Instruct-1M. Low-rank adaptation is used to reduce the cost of policy optimization. The planned LoRA configuration uses a rank of 8 and a LoRA alpha of 32. A fixed copy of the initial policy backbone is used as the reference model for calculating the KL divergence penalty. The number of generated responses per context is treated as the GRPO group size. The exact learning rate, batch size, number of epochs, and number of GRPO training client cases will be finalized during implementation based on available computational resources.

The reward model and policy optimization stages are separated to make the experimental comparison clearer. The reward model training stage evaluates whether the system can learn to approximate therapeutic reward scores from lookahead data. The GRPO stage then evaluates whether this learned reward signal can improve the therapeutic response generation policy across different policy model backbones.

4.2 Experimental Results

This section presents the current experimental results and describes how the final results will be reported after the full AIM-RGL implementation and evaluation are completed. At the current stage, the available results are preliminary baseline results comparing CBT and FLASH under easy, normal, and hard simulated client settings. The final version of this section will include the complete comparison between CBT, CAMEL, FLASH, and AIM-RGL, together with lookahead horizon, reward model backbone, and policy model backbone comparisons.

4.2.1 Therapeutic Response Quality Results

The preliminary evaluation compares CBT and FLASH using therapist skill dimensions and broader therapeutic quality metrics. The therapist skill dimensions include approach, discovery, focus, strategy, understanding, and interpersonal collaboration. Table 4.2 summarizes the preliminary therapist skill results across easy, medium, and hard simulated client levels.

Table 4.2

Preliminary therapist skill results for CBT and FLASH.

Level	System	Approach	Discovery	Focus	Strategy	Understanding	Collaboration
Easy	CBT	4.00	4.00	3.60	4.00	4.80	4.00
Easy	FLASH	4.00	3.80	3.60	4.00	4.60	3.60
Medium	CBT	4.00	4.00	3.20	4.00	4.40	4.00
Medium	FLASH	4.00	3.40	3.40	4.00	4.80	3.40
Hard	CBT	4.00	3.80	3.40	4.00	4.20	3.20
Hard	FLASH	4.00	3.40	3.00	3.80	5.00	3.30

The preliminary results show that CBT obtains stronger scores than FLASH on several therapist skill dimensions, especially discovery and collaboration. CBT also remains relatively stable across difficulty levels. FLASH shows stronger understanding scores in medium and hard settings, but its discovery, focus, and collaboration scores are lower in several conditions. This suggests that FLASH may generate responses that appear understanding, but may provide less structured therapeutic progression in more difficult simulated client conditions.

Table 4.3 summarizes the preliminary therapeutic quality and alliance results. The evaluation includes CTRS, WAI, MITI, and alliance scores across easy, normal, and hard simulated client settings.

Table 4.3

Preliminary therapeutic skills and alliance results.

Level	System	CTRS	WAI	MITI	Alliance
Easy	CBT	3.73	3.67	4.22	3.84
Easy	FLASH	3.47	3.43	4.24	3.64
Normal	CBT	3.65	3.55	4.04	3.79
Normal	FLASH	2.77	3.23	3.92	3.27
Hard	CBT	3.61	3.71	3.88	3.77
Hard	FLASH	2.82	3.06	3.80	3.24

The preliminary results indicate that CBT performs better than FLASH on CTRS, WAI, and alliance across all difficulty levels. FLASH obtains a slightly higher MITI score in the easy condition, but CBT performs better on MITI in the normal and hard conditions. This suggests that CBT provides stronger structured therapeutic competence and alliance in the current preliminary evaluation, while FLASH may preserve some MI-related qualities in easier settings. However, these findings should be interpreted only as preliminary baseline results because the full AIM-RGL comparison has not yet been completed.

4.2.2 Ablation and Model Comparison Results

The ablation and model comparison results will be added after the full AIM-RGL experiment is completed. This part of the experiment will examine whether reward-guided lookahead search improves therapeutic response quality compared with the baseline systems. It will also analyze how different lookahead horizons affect response quality.

The lookahead comparison will evaluate one-step, two-step, and three-step lookahead settings. The purpose of this comparison is to determine whether deeper lookahead improves therapeutic response selection by considering more future client reactions, or whether the benefit becomes limited after a certain depth. The final results will be reported using the same therapeutic evaluation metrics, including CTRS, WAI, MITI, alliance, change talk, and sustain talk.

Table 4.4

Planned lookahead horizon comparison.

Lookahead Horizon	CTRS	WAI	MITI	Change Talk	Sustain Talk
One-step lookahead	–	–	–	–	–
Two-step lookahead	–	–	–	–	–
Three-step lookahead	–	–	–	–	–

The reward model backbone comparison will evaluate DeBERTa-v3-large, ModernBERT-large, and RoBERTa-large. Since the reward model predicts a scalar therapeutic reward score, the comparison will use regression and ranking-based metrics such as mean squared error, mean absolute error, Pearson correlation, and Spearman rank correlation.

The policy model backbone comparison will evaluate LLaMA-3.1-8B-Instruct and Qwen-2.5-7B-Instruct-1M after policy optimization using the trained reward model.

Table 4.5

Planned reward model backbone comparison.

Reward Model Backbone	MSE	MAE	Pearson r	Spearman ρ
DeBERTa-v3-large	–	–	–	–
ModernBERT-large	–	–	–	–
RoBERTa-large	–	–	–	–

This comparison will show whether the quality of MI-CBT response generation depends on the underlying policy model backbone.

Table 4.6

Planned policy model backbone comparison.

Policy Backbone	CTRS	WAI	MITI	Change Talk	Sustain Talk
LLaMA-3.1-8B-Instruct	–	–	–	–	–
Qwen-2.5-7B-Instruct-1M	–	–	–	–	–

4.2.3 Summary of Findings

The current preliminary results provide an initial baseline comparison between CBT and FLASH. CBT shows stronger overall performance in structured therapist skill quality, CTRS, WAI, and alliance across easy, normal, and hard simulated client settings. FLASH shows competitive MI-related performance in the easy setting, but its scores decrease more clearly in normal and hard simulated client conditions.

These preliminary findings support the need for a response selection mechanism that can balance structured CBT competence with MI-consistent readiness handling. The final AIM-RGL evaluation will examine whether reward-guided lookahead search can improve this balance by comparing candidate MI-CBT responses according to their expected near-future therapeutic effects. The final summary will be updated after AIM-RGL, lookahead horizon, reward model backbone, and policy backbone results are completed.

CHAPTER 5

CONCLUSION

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APPENDICES

APPENDIX A

TITLE